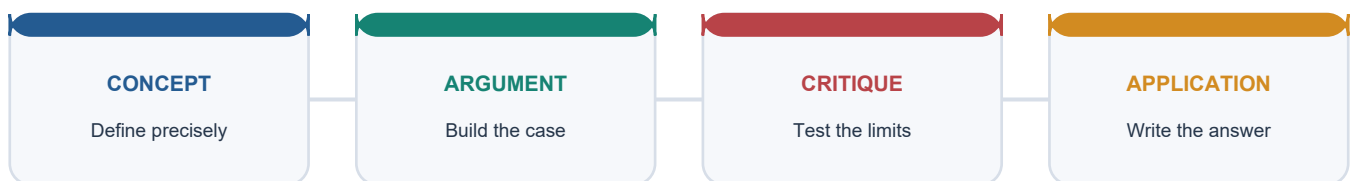


COMPLETE LEARNING SESSION

Climate Economics, Green Finance and Circular Economy - Solved Practice Workbook

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing

LEARNING PATH



Deterministic fallback visual: move from precise concepts through argument and critique to exam application.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

Climate Economics, Green Finance and Circular Economy - Solved Practice Workbook

Exactly 32 original questions appear before PYQs. Correct answers rotate A -> B -> C -> D eight times.

BASIC MCQS / REMEDIATION

MCQ 1

Why does an unpriced greenhouse-gas emission create a market failure?

A. Its social marginal cost can exceed the private cost faced by the emitter B. It always lowers the emitter's private profit C. It is necessarily excluded from national accounts D. It affects only the country where it is emitted

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: uncompensated external damage creates the private-social cost wedge.
- **B:** Incorrect: Incorrect: emitting activity may remain privately profitable.
- **C:** Incorrect: Incorrect: accounting inclusion does not internalise external damage.
- **D:** Incorrect: Incorrect: greenhouse gases create cross-border and intergenerational effects.

Examiner trap 1: Do not confuse an external cost with an accounting omission.

MCQ 2

Which feature most directly generates international free riding in climate mitigation?

A. Carbon dioxide has no measurable atmospheric effect B. Countries can benefit from others' mitigation while avoiding part of the domestic cost C. Every country faces identical mitigation cost D. The Paris Agreement sets one compulsory tax rate

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: physical effects are measurable even with uncertainty.
- **B:** Correct: Correct: diffuse benefits and nationally borne costs create the incentive.
- **C:** Incorrect: Incorrect: costs and capabilities differ.
- **D:** Incorrect: Incorrect: the Paris system does not impose one global tax.

Examiner trap 2: Public-good benefits make cooperation difficult, not unnecessary.

MCQ 3

Which sequence correctly distinguishes climate responses?

A. Adaptation cuts emissions; mitigation compensates damage; loss and damage builds resilience B. Loss and damage is another name for mitigation C. Mitigation lowers future hazard, adaptation lowers vulnerability, and loss and damage addresses residual impacts D. Adaptation makes mitigation redundant

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: the functions are reversed and mixed.
- **B:** Incorrect: Incorrect: residual impacts are distinct from emission reduction.
- **C:** Correct: Correct: the sequence follows the risk chain.
- **D:** Incorrect: Incorrect: both responses remain necessary.

Examiner trap 3: Residual harm can remain even after feasible adaptation.

MCQ 4

The Social Cost of Carbon is best described as:

A. The annual market price of a CCTS certificate B. A statutory carbon-tax rate fixed by Parliament C. The cost of producing one tonne of coal D. A discounted estimate of marginal damage from one additional tonne of CO₂-equivalent emissions

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: a traded price reflects certificate supply and demand.
- **B:** Incorrect: Incorrect: a tax may be informed by SCC but is legally set.
- **C:** Incorrect: Incorrect: production cost is not social climate damage.
- **D:** Correct: Correct: SCC is a model-based appraisal value.

Examiner trap 4: Never quote one universal SCC without a dated model and assumptions.

MCQ 5

A carbon tax primarily provides:

A. Price certainty per covered unit while emissions quantity adjusts B. A fixed aggregate quantity with a fluctuating price C. Project credits without a compliance base D. A renewable-electricity attribute

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: government fixes the tax rate.
- **B:** Incorrect: Incorrect: that describes cap-and-trade.
- **C:** Incorrect: Incorrect: that describes an offset mechanism.
- **D:** Incorrect: Incorrect: that describes an REC.

Examiner trap 5: Price certainty is not emissions-quantity certainty.

MCQ 6

Under cap-and-trade, the regulator ordinarily fixes:

A. The exact profit of each firm B. The aggregate allowance quantity or cap C. The Social Cost of Carbon D. The output of every covered plant

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: firms retain commercial outcomes.
- **B:** Correct: Correct: allowance scarcity creates the traded price.
- **C:** Incorrect: Incorrect: SCC is an appraisal estimate.
- **D:** Incorrect: Incorrect: a cap constrains emissions, not necessarily output.

Examiner trap 6: Allowance allocation does not alter the environmental meaning of the cap.

MCQ 7

A baseline-and-credit system differs from an absolute cap because it:

A. Cannot use trading B. Always fixes a statutory tax C. Issues credits relative to a performance benchmark, often linked to output D. Requires every project to be outside regulated sectors

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: credits can be traded.

- **B:** Incorrect: Incorrect: no tax is inherent.
- **C:** Correct: Correct: benchmark performance determines credits or shortfalls.
- **D:** Incorrect: Incorrect: that restriction belongs to particular offset designs, not the definition.

Examiner trap 7: Intensity improvement may coexist with higher absolute emissions.

MCQ 8

Which integrity condition asks whether a credited reduction would have occurred anyway?

A. Permanence B. Leakage C. Corresponding adjustment D. Additionality

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: permanence concerns reversal.
- **B:** Incorrect: Incorrect: leakage concerns displaced emissions.
- **C:** Incorrect: Incorrect: corresponding adjustment addresses international double claiming.
- **D:** Correct: Correct: additionality tests the counterfactual.

Examiner trap 8: A registered project is not automatically additional.

MCQ 9

PAT certificates reward:

A. Verified overachievement of specified energy-consumption targets by designated consumers B. Every tonne of renewable electricity generated C. Any voluntary tree plantation D. Ownership of a sovereign green bond

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: PAT is an energy-efficiency intensity mechanism.
- **B:** Incorrect: Incorrect: renewable attributes belong to REC rules.
- **C:** Incorrect: Incorrect: offsets require approved methodologies.
- **D:** Incorrect: Incorrect: bond ownership is unrelated to PAT compliance.

Examiner trap 9: PAT is not an economy-wide absolute carbon cap.

MCQ 10

A Renewable Energy Certificate primarily represents:

A. A bank's green-deposit liability B. The environmental attribute of eligible renewable electricity C. One unit of avoided plastic waste D. A carbon tax payment receipt

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: green deposits are banking liabilities.
- **B:** Correct: Correct: electricity and its renewable attribute can be accounted separately.
- **C:** Incorrect: Incorrect: plastic EPR has its own certificates.
- **D:** Incorrect: Incorrect: a tax receipt is not an REC.

Examiner trap 10: REC, ESCert and Carbon Credit Certificate are distinct instruments.

MCQ 11

Which statement about India's CCTS is correct?

- A. SEBI administers it as a mutual-fund product B. It is a universal absolute cap covering the whole economy C. It contains a compliance mechanism and a project-based offset mechanism under the Energy Conservation framework D. It is the EU's border adjustment

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: BEE administers the scheme and CERC oversees trading.
- **B:** Incorrect: Incorrect: coverage is phased and targets are intensity-based.
- **C:** Correct: Correct: the 2023 scheme and amendment establish both arms.
- **D:** Incorrect: Incorrect: CBAM is an EU import measure.

Examiner trap 11: Legal creation does not prove mature market liquidity.

MCQ 12

As of the 10 September 2026 cutoff, the safest statement on CCTS market status is:

- A. A single nationwide carbon price had replaced all energy regulation B. All proposed sectors had final targets C. Trading volume was known even without an official record D. Notified obligations and market infrastructure were advancing, but no unsupported price or mature-trading claim should be made

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: India retains multiple policy instruments.
- **B:** Incorrect: Incorrect: iron-and-steel targets were still draft in the checked evidence.
- **C:** Incorrect: Incorrect: price and volume require official transaction evidence.
- **D:** Correct: Correct: it distinguishes legal and operational stages.

Examiner trap 12: Do not convert registry or accreditation readiness into proven exchange trading.

MCQ 13

UNFCCC's broad climate-finance concept includes:

- A. Public, private and alternative finance supporting mitigation and adaptation B. Only grants transferred between governments C. Only money raised through green bonds D. Every loan to an energy company

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: sources and channels are broad but purpose matters.
- **B:** Incorrect: Incorrect: private and alternative sources also count.
- **C:** Incorrect: Incorrect: bonds are one instrument.
- **D:** Incorrect: Incorrect: eligibility and climate purpose must be established.

Examiner trap 13: Climate co-benefit and climate-finance classification are not automatically identical.

MCQ 14

Mobilised private climate finance is best separated from provided public finance because:

A. Private capital is never climate finance B. Mobilisation depends on attribution to a public intervention and can be double counted C. Public finance cannot be concessional D. Only disbursement matters in all reports

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: private finance can support climate action.
- **B:** Correct: Correct: attribution and overlap must be controlled.
- **C:** Incorrect: Incorrect: public finance can include grants and concessional loans.
- **D:** Incorrect: Incorrect: commitments and mobilisation are analytically relevant but distinct.

Examiner trap 14: Do not add leveraged project value to public finance without checking methodology.

MCQ 15

Blended finance is most defensible when it:

A. Guarantees every project loss B. Provides permanent subsidies to incumbents C. Uses targeted public risk absorption to overcome a specific barrier and mobilise additional private capital D. Replaces environmental regulation

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: unlimited guarantees socialise losses.
- **B:** Incorrect: Incorrect: permanent rents undermine additionality.
- **C:** Correct: Correct: barrier, leverage and exit conditions define sound blending.
- **D:** Incorrect: Incorrect: finance cannot substitute for standards and enforcement.

Examiner trap 15: Leverage is not proof of welfare or environmental additionality.

MCQ 16

Why is adaptation often difficult to finance commercially?

A. Adaptation never produces benefits B. All adaptation assets are illegal collateral C. Mitigation projects receive no finance D. Many benefits are diffuse avoided losses without a direct revenue stream

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: social returns can be high.
- **B:** Incorrect: Incorrect: collateral varies by project.
- **C:** Incorrect: Incorrect: mitigation can attract substantial finance.
- **D:** Correct: Correct: public-good benefits weaken cash-flow capture.

Examiner trap 16: Low bankability can coexist with high social value.

MCQ 17

A green bond is principally:

A. Debt with proceeds earmarked and tracked for eligible environmental uses B. Equity in a renewable company C. A grant with no repayment obligation D. A guarantee of verified emissions reduction

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: use-of-proceeds rules distinguish the label.
- **B:** Incorrect: Incorrect: a bond remains debt.

- **C:** Incorrect: Incorrect: principal and interest remain due.
- **D:** Incorrect: Incorrect: impact requires separate evidence.

Examiner trap 17: Green use of proceeds does not eliminate issuer credit risk.

MCQ 18

India's FY 2023-24 sovereign green-bond allocation report should be read as:

A. Rs 20,785.60 crore of green-bond issuance B. Rs 20,000 crore issuance with Rs 20,785.60 crore eligible transfers, including Rs 785.60 crore from general revenue C. A verified carbon-credit market price D. A private corporate bond report

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: the excess transfer was not bond proceeds.
- **B:** Correct: Correct: it reconciles issuance and budgetary top-up.
- **C:** Incorrect: Incorrect: allocation reporting does not set carbon prices.
- **D:** Incorrect: Incorrect: the issuer is the Government of India.

Examiner trap 18: Eligible expenditure can exceed issuance without increasing bond proceeds.

MCQ 19

A claimed greenium is most credible when:

A. Any green bond yield is compared with any treasury bill B. The issuer reports a high environmental score C. Yield is compared with a conventional bond matched for issuer, maturity, liquidity and date D. The bond finances renewable energy

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: unmatched instruments make the spread meaningless.
- **B:** Incorrect: Incorrect: scores do not isolate pricing.
- **C:** Correct: Correct: matched comparison reduces confounding.
- **D:** Incorrect: Incorrect: project type alone does not measure a yield premium.

Examiner trap 19: A greenium is a financing-price observation, not an environmental outcome.

MCQ 20

Which statement about RBI-regulated green deposits is correct?

A. They are equity securities B. They must offer a higher interest rate C. They are exempt from normal repayment obligations D. They are fixed-period deposits whose proceeds are earmarked under a board-approved financing framework

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: they remain deposits.
- **B:** Incorrect: Incorrect: the framework does not mandate a premium rate.
- **C:** Incorrect: Incorrect: the institution remains liable to the depositor.
- **D:** Correct: Correct: earmarking, review and reporting govern the green use.

Examiner trap 20: The green label changes proceeds governance, not the basic depositor claim.

MCQ 21

Flood damage to a borrower's factory is an example of:

A. Physical climate risk transmitting into credit risk B. Only transition risk C. Only reputational risk D. A carbon-market offset

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: hazard affects operations, collateral and repayment.
- **B:** Incorrect: Incorrect: transition risk arises from economic adjustment.
- **C:** Incorrect: Incorrect: reputation may be secondary but does not define the event.
- **D:** Incorrect: Incorrect: a loss is not an offset.

Examiner trap 21: Hazard becomes financial risk through exposure and vulnerability.

MCQ 22

A coal plant losing value after stricter policy and cheaper alternatives illustrates:

A. Acute physical risk B. Transition risk and possible asset stranding C. Adaptation finance D. Natural-capital appreciation

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: no weather event is necessary.
- **B:** Correct: Correct: policy and technology can shorten economic life.
- **C:** Incorrect: Incorrect: adaptation addresses climate impacts.
- **D:** Incorrect: Incorrect: the asset is being impaired.

Examiner trap 22: Transition risk can rise when adjustment is delayed and abrupt.

MCQ 23

Liability climate risk most directly involves:

A. Normal depreciation B. A rainfall forecast C. Legal claims or compensation duties arising from climate harm or misleading disclosure D. A renewable purchase obligation

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: ordinary depreciation need not involve climate liability.
- **B:** Incorrect: Incorrect: a forecast alone is not a claim.
- **C:** Correct: Correct: litigation and legal responsibility define liability risk.
- **D:** Incorrect: Incorrect: RPO is a compliance instrument.

Examiner trap 23: Liability, physical and transition risks may interact but should be named separately.

MCQ 24

Climate scenario analysis should be interpreted as:

A. A certain forecast of next year's losses B. A replacement for all credit appraisal C. A promise that no severe event will occur D. A conditional exploration of portfolio resilience under stated pathways and assumptions

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: scenarios are not point forecasts.
- **B:** Incorrect: Incorrect: they complement risk management.

- **C:** Incorrect: Incorrect: they test rather than eliminate risk.
- **D:** Correct: Correct: assumptions and pathways define interpretation.

Examiner trap 24: A severe scenario is not the regulator's prediction.

MCQ 25

Who mandates BRSR for covered listed companies?

A. SEBI B. RBI C. BEE D. CPCB

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: SEBI regulates listed-entity disclosure.
- **B:** Incorrect: Incorrect: RBI regulates specified financial entities and deposits.
- **C:** Incorrect: Incorrect: BEE administers energy efficiency and CCTS.
- **D:** Incorrect: Incorrect: CPCB implements pollution and waste functions.

Examiner trap 25: Do not attribute securities disclosure to the banking regulator.

MCQ 26

Under SEBI's March 2025 circular, value-chain ESG disclosure for the top 250 listed entities is:

A. Mandatory from FY 2023-24 without exception B. Voluntary from FY 2025-26 C. Prohibited until FY 2027-28 D. Identical to BRSR Core assessment for the top 500

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: the 2025 revision changed the earlier path.
- **B:** Correct: Correct: value-chain disclosure became voluntary from FY 2025-26.
- **C:** Incorrect: Incorrect: voluntary disclosure is permitted.
- **D:** Incorrect: Incorrect: entity coverage and obligation differ.

Examiner trap 26: Separate BRSR, BRSR Core and value-chain requirements.

MCQ 27

As of the cutoff, India's national climate-finance taxonomy was:

A. A final statute enacted in 2022 B. A SEBI mutual-fund category C. A DEA draft framework released on 7 May 2025, with no final framework identified D. The same as the EU taxonomy

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: the Union Budget announcement came later.
- **B:** Incorrect: Incorrect: it is a whole-economy classification initiative.
- **C:** Correct: Correct: draft status must be retained.
- **D:** Incorrect: Incorrect: interoperability does not mean identity.

Examiner trap 27: Never upgrade consultation text into final binding criteria.

MCQ 28

Which chain best tests greenwashing?

A. Label alone B. High subscription alone C. Issuer reputation alone D. Eligibility, allocation, output, outcome and additionality with verification

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: labels are assertions.
- **B:** Incorrect: Incorrect: investor demand measures finance, not impact.
- **C:** Incorrect: Incorrect: reputation cannot replace evidence.
- **D:** Correct: Correct: the chain follows funds to environmental consequence.

Examiner trap 28: External review reduces but does not eliminate model and conflict risk.

MCQ 29

A just transition focuses primarily on:

A. Workers, regions, consumers and access affected by structural decarbonisation B. Protecting every incumbent asset indefinitely C. Replacing mitigation with compensation D. Guaranteeing identical local outcomes

Answer: A.

Option-wise explanation:

- **A:** Correct: Correct: distribution and adjustment are central.
- **B:** Incorrect: Incorrect: indefinite protection blocks transition.
- **C:** Incorrect: Incorrect: support complements mitigation.
- **D:** Incorrect: Incorrect: places differ in exposure and capability.

Examiner trap 29: Justice concerns participation and future opportunity, not only cash compensation.

MCQ 30

Extended Producer Responsibility means:

A. Government physically collects every product B. Producers bear specified responsibility for post-consumer collection, recycling or sound management C. Consumers have no duties D. All waste certificates are carbon credits

Answer: B.

Option-wise explanation:

- **A:** Incorrect: Incorrect: producers may meet obligations through authorised systems.
- **B:** Correct: Correct: lifecycle responsibility shifts upstream.
- **C:** Incorrect: Incorrect: segregation and return duties may remain.
- **D:** Incorrect: Incorrect: EPR certificates belong to waste-specific regimes.

Examiner trap 30: EPR obligations and certificate rules vary by waste stream.

MCQ 31

Life-cycle assessment prevents which error?

A. Measuring any environmental impact B. Using a functional unit C. Calling zero operational emissions a zero-impact lifecycle without examining production and disposal D. Comparing technology options

Answer: C.

Option-wise explanation:

- **A:** Incorrect: Incorrect: LCA measures multiple impacts.
- **B:** Incorrect: Incorrect: a functional unit is essential.

- **C:** Correct: Correct: impacts can shift across stages.
- **D:** Incorrect: Incorrect: comparison is a core use when boundaries align.

Examiner trap 31: Always disclose system boundary and electricity/material assumptions.

MCQ 32

Which package best addresses climate and circular-economy market failures?

A. Only voluntary disclosure B. Only a recycling target C. Only a carbon tax D. Carbon pricing, standards, innovation, public investment, circular design and social protection

Answer: D.

Option-wise explanation:

- **A:** Incorrect: Incorrect: disclosure alone does not change every incentive.
- **B:** Incorrect: Incorrect: prevention and repair precede recycling.
- **C:** Incorrect: Incorrect: tax cannot supply networks or protect vulnerable groups alone.
- **D:** Correct: Correct: the package maps multiple failures to complementary tools.

Examiner trap 32: Instrument quantity is not quality; check coherence and outcomes.

PYQS AND ANSWER PRACTICE

VERIFIED MAINS PYQ 1 - 2025 GS-III - 10 MARKS

Question: How can India achieve energy independence through clean technology by 2047? How can biotechnology play a crucial role in this endeavour? Answer in 150 words.

Model solution: Energy independence means reducing exposure to imported fossil fuels and concentrated clean-technology supply chains while preserving reliable, affordable energy. India needs renewable power, storage, stronger grids, electrified transport, efficient buildings and industry, green hydrogen for hard-to-abate uses, domestic critical-component capability and diversified mineral partnerships. Demand management and circular recovery of batteries and metals reduce import intensity. Biotechnology supports second-generation biofuels from residues, compressed biogas, algal fuels, engineered enzymes, waste-to-energy and climate-resilient biomass. It can also improve process efficiency and substitute fossil feedstocks in chemicals. Public research, demonstration finance, standards and procurement should support learning without protecting inefficient firms indefinitely. Land, water, food-security, biosafety and lifecycle-emission safeguards are essential. Energy independence is therefore diversified resilience, not autarky or a mere renewable-capacity target.

VERIFIED MAINS PYQ 2 - 2025 GS-III - 10 MARKS

Question: What is Carbon Capture, Utilization and Storage (CCUS)? What is the potential role of CCUS in tackling climate change? Answer in 150 words.

Model solution: CCUS separates carbon dioxide from concentrated industrial or energy streams, compresses and transports it, uses it in products or processes where viable, or injects it into suitable geological formations for long-term storage. It can reduce residual emissions in cement, steel, chemicals and refineries where process emissions or high-temperature heat are difficult to eliminate. It may also support low-carbon hydrogen and, when paired with sustainable biomass or direct air capture, carbon removal. However, capture consumes energy, raises cost and requires transport networks, storage characterisation, monitoring and liability rules. Utilisation may delay rather than permanently avoid release, and enhanced fossil extraction can undermine benefits. CCUS should therefore complement efficiency, electrification, renewable energy and material substitution, not preserve avoidable fossil dependence.

VERIFIED MAINS PYQ 3 - 2025 GS-III - 15 MARKS

Question: Write a review on India's climate commitments under the Paris Agreement (2015) and mention how these have been further strengthened in COP26 (2021). In this direction, how has the first Nationally Determined Contribution intended by India been updated in 2022? Answer in 250 words.

Model solution: India's first NDC under the Paris Agreement combined development with differentiated responsibility. Its 2030 goals included reducing GDP emissions intensity by 33-35 per cent from 2005, achieving 40 per cent cumulative installed electric-power capacity from non-fossil sources with support, and creating an additional 2.5-3 billion tonnes of CO₂-equivalent forest and tree carbon sink. At COP26 in 2021, India announced the Panchamrit political package: 500 GW non-fossil capacity, about 50 per cent energy requirements from renewables, one billion tonnes reduction in projected emissions to 2030, 45 per cent emissions-intensity reduction, and net zero by 2070. The formal 2022 NDC update incorporated a 45 per cent emissions-intensity reduction and about 50 per cent cumulative installed electric-power capacity from non-fossil sources by 2030, along with LiFE. Political announcements and formally communicated NDC text should remain distinct. Delivery requires grids, storage, finance, industrial decarbonisation, adaptation and a just transition. Progress should be assessed against dated official inventories and capacity data, not promotional claims.

VERIFIED OBJECTIVE PYQ 1 - 2025 PRELIMS GS-I Q4

Question: Consider the following statements: I. The Reserve Bank of India mandates all the listed companies in India to submit a Business Responsibility and Sustainability Report (BRSR). II. In India, a company submitting a BRSR makes disclosures in the report that are largely non-financial in nature.

Which of the statements given above is/are correct?

A. I only B. II only C. Both I and II D. Neither I nor II

Official Set-A key: B.

Solution: Statement I is incorrect because SEBI, not RBI, mandates BRSR within the listed-company perimeter. Statement II is correct because BRSR principally reports environmental, social and governance information. Do not confuse ordinary BRSR coverage with the phased BRSR Core assessment/assurance glide path.

VERIFIED OBJECTIVE PYQ 2 - 2025 PRELIMS GS-I Q9

Question: Consider the following statements: I. Circular economy reduces the emissions of greenhouse gases. II. Circular economy reduces the use of raw materials as inputs. III. Circular economy reduces wastage in the production process.

Which one of the following is correct in respect of the above statements?

A. Both Statement II and Statement III are correct and both of them explain Statement I B. Both Statement II and Statement III are correct but only one of them explains Statement I C. Only one of the Statements II and III is correct and that explains Statement I D. Neither Statement II nor Statement III is correct

Official Set-A key: A.

Solution: Lower virgin-material extraction and lower production waste can both reduce energy use and greenhouse-gas emissions, so Statements II and III are correct and both explain Statement I. The conclusion is a causal mechanism, not a claim that every recycling process has zero lifecycle emissions.

KEY-NEUTRAL OBJECTIVE ROUTES

- **2020 Prelims Q85:** Social Cost of Carbon as monetary valuation of incremental emissions damage. The official key is not held locally, so no answer letter is inferred.
- **2022 Prelims Q80:** Greenwashing as false or misleading environmental claims. The official key is not held locally, so no answer letter is inferred.
- **2026 Prelims Q92:** Sustainability bonds and combined environmental/social financing. The locally held key is provisional, so no answer letter is asserted.

ORIGINAL MAINS PRACTICE

MAINS 1

Question: Distinguish a carbon tax, cap-and-trade and baseline-and-credit system. Answer in 150 words.

Model answer:

A carbon tax sets a statutory price per covered tonne; firms abate when doing so costs less, leaving emissions quantity uncertain. Cap-and-trade fixes an aggregate allowance quantity; trading discovers the price, leaving compliance cost uncertain. Baseline-and-credit instead compares performance with a benchmark, often emissions intensity per unit of output. Beating the benchmark earns credits; excess intensity creates a purchase or surrender obligation. India's CCTS compliance arm follows this intensity-linked logic, not an economy-wide absolute cap. Each instrument can align marginal abatement incentives, but integrity requires credible coverage, measurement and enforcement. Tax design must address revenue use; cap-and-trade must prevent overallocation and volatility; baseline-credit must prevent weak benchmarks and rising absolute emissions. Choice should reflect administrative capacity, distribution, political acceptability across affected sectors, and whether policy prioritises price or quantity certainty.

Native-body ceiling: 150 words. Independently measured in validation; heading and metadata excluded.

MAINS 2

Question: Why does adaptation face a financeability gap? Answer in 150 words.

Model answer:

Adaptation reduces exposure and vulnerability to climate hazards, but many benefits appear as avoided future losses rather than saleable cash flows. Flood barriers, heat-health systems, climate information and watershed resilience protect whole communities, making exclusion difficult and private revenue uncertain. Benefits are local, long-dated and dependent on uncertain hazard pathways, while vulnerable borrowers may lack collateral or capacity to service debt. Commercial lenders therefore underprovide finance despite high social returns. Public budgets and grants should fund core public goods; concessional loans, guarantees and blended structures can support revenue-bearing components; insurance and catastrophe instruments can spread residual risk. Project preparation, local data and measurable resilience indicators improve bankability. Yet leverage must not be mistaken for additionality, and debt should not shift climate costs onto already vulnerable communities. Adaptation finance requires public-value appraisal alongside conventional financial return.

Native-body ceiling: 150 words. Independently measured in validation; heading and metadata excluded.

MAINS 3

Question: Evaluate India's sovereign green bonds and green deposits as climate-finance instruments. Answer in 250 words.

Model answer:

Sovereign green bonds and green deposits both earmark funds, but their issuers and liabilities differ. India's sovereign green-bond framework, approved on 9 November 2022, governs central-government borrowing for eligible public green expenditure. Proceeds enter the Consolidated Fund, while the Green Finance Working Committee selects expenditure and the Public Debt Management Cell tracks allocation. The FY 2023-24 report records Rs 20,000 crore issued and Rs 20,785.60 crore transferred to eligible schemes, with the excess Rs 785.60 crore funded from general revenue. Green deposits are fixed-period liabilities of covered banks or deposit-taking NBFCs/HFCs. RBI's 28 November 2025 directions require board policy, a financing framework, external review, annual allocation assurance and impact assessment. Both instruments can widen the investor or depositor base and improve traceability. However, neither label guarantees additional emissions reduction, resilience or cheaper finance. Sovereign allocation is not impact, a greenium requires a matched conventional benchmark, and deposit repayment remains the regulated entity's obligation. Credibility requires final taxonomy criteria, transparent exclusions, project-level tracking, independent review and outcome reporting.

Native-body ceiling: 250 words. Independently measured in validation; heading and metadata excluded.

MAINS 4

Question: How can climate risk threaten financial stability? Examine RBI's developing response. Answer in 250 words.

Model answer:

Climate risk reaches finance through physical, transition and liability channels. Floods, heat and chronic water stress damage assets, disrupt production and weaken collateral and repayment. Policy, technology and preference shifts can strand carbon-intensive assets, while litigation or misleading disclosure can create compensation and reputation costs. Concentrated exposures may then generate correlated defaults, insurance withdrawal, fire sales and operational disruption, turning institution-level losses into systemic feedback. RBI's approach separates facilitation from prudential management. Its green-deposit framework channels funds under use-of-proceeds controls. The draft Disclosure Framework of 28 February 2024 proposes governance, strategy, risk-management, and metrics-and-targets reporting for specified regulated entities. A March 2025 RBI speech stated that the disclosure guidelines were being finalised, scenario-analysis and stress-testing guidance was under development, and RB-CRIS was intended to address data gaps. By the 10 September 2026 cutoff, no final notification for that separate disclosure draft was identified. RBI should combine comparable data, sector-geography mapping, scenario analysis, supervisory dialogue and proportionate capital planning. Scenarios must remain conditional exercises, not forecasts, and disclosure cannot substitute for risk mitigation.

Native-body ceiling: 250 words. Independently measured in validation; heading and metadata excluded.

MAINS 5

Question: Explain how EPR and circular-economy design can reduce material use and emissions. Identify implementation risks. Answer in 250 words.

Model answer:

A circular economy seeks to prevent waste and preserve embedded value through refusal, redesign, reduction, reuse, repair, refurbishment and remanufacture before residual recycling. Lower virgin-material extraction and production waste can reduce energy use and greenhouse-gas emissions, the causal link recognised in the 2025 UPSC objective key. Extended Producer Responsibility translates lifecycle thinking into enforceable obligations: producers must meet specified post-consumer collection, recycling or sound-management duties through registered systems. India has category-specific regimes for plastic packaging, batteries, e-waste, waste tyres and used oil, each with its own definitions, targets and portal. EPR can finance reverse logistics, create demand for secondary material and incentivise design for durability and recyclability. Yet certificate markets can reward paper compliance if baselines, mass balance and processor capacity are weak. Informal collectors may lose livelihoods or material access if integration is ignored. Recycling can itself use energy and create pollution, while cheaper secondary inputs may induce rebound. Policy should prioritise waste prevention, eco-design, repair rights, recycled-content standards, traceable certificates, independent audits, worker integration and lifecycle assessment. Success requires lower absolute material throughput and pollution, not only more registrations or certificates.

Native-body ceiling: 250 words. Independently measured in validation; heading and metadata excluded.

MAINS 6

Question: Design a balanced climate-policy mix for India that combines efficiency, finance and justice. Answer in 250 words.

Model answer:

India needs a portfolio because climate change reflects several failures: unpriced emissions, technology spillovers, infrastructure gaps, information asymmetry and unequal adjustment capacity. A credible carbon signal can reward lower-emission choices, but CCTS intensity benchmarks require strong MRV and should be read alongside absolute emissions. Efficiency standards and EPR address products and materials that are difficult to price fully. Public investment in grids, storage, transport, water and resilient cities supplies network and public goods; research, demonstration and procurement support learning in emerging technologies. Green bonds, deposits and blended finance can mobilise capital when taxonomy, allocation, assurance and impact reporting prevent greenwashing. Adaptation needs grants and concessional finance because many benefits are diffuse avoided losses. Distribution

must be designed, not appended: recycle revenues progressively, preserve lifeline energy access, fund skills and social protection, diversify coal-dependent regions and include informal waste workers. SEBI and RBI disclosures should improve information, while environmental regulators enforce outcomes. CBAM-related competitiveness requires credible embedded-emissions data, not indiscriminate protection. The policy dashboard should track absolute emissions, intensity, resilience, material throughput, jobs, affordability, fiscal risk and additionality. The objective is an orderly, innovative and just transition rather than maximum instrument count.

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